

Note: When you ... marks):
 Q1) Circle the correct answers, and then move these ... marks):
 1) Three forces are in equilibrium depending on the shown table, the magnitude of (θ_3) according (X°) is:

- A) 84.1° B) 95.9° C) 255.5° D) 275°

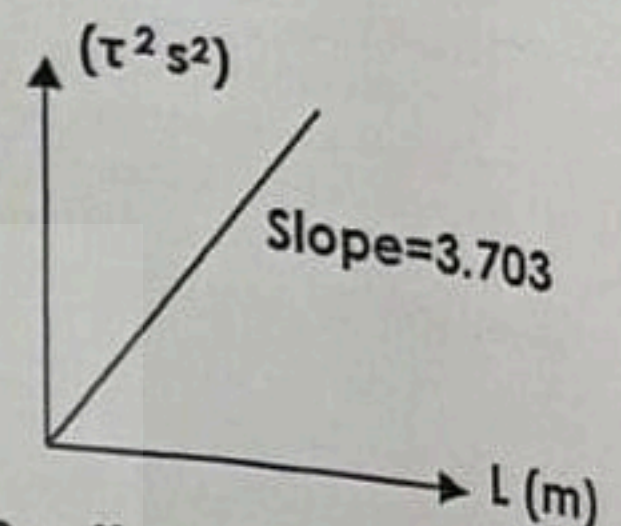
F	θ	F_x	F_y
F_1	30°	6.9282	4
F_2	130°	-4.4995	5.3623
F_3			

2) In uniform circular motion experiment, the relation between the radius (r) of the circle and the periodic time (τ) is:

- A) Inversely proportional B) Directly proportional C) no relation

3) In simple pendulum experiment, the figure below shows the relationship between the square of periodic time (τ^2) versus the length of the cord (L): The acceleration of gravity g in (m/s^2) is:

- A) 10.65 B) 10.11 C) 9.62 D) 9.17

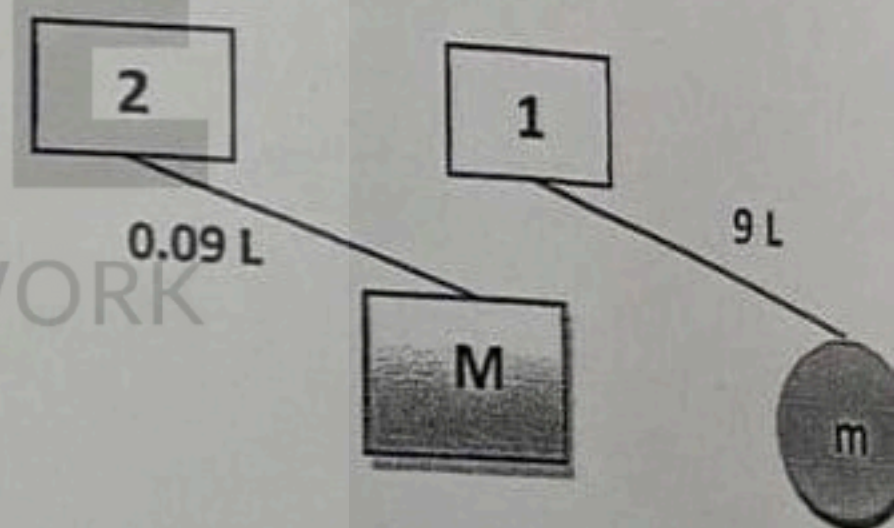


4) In uniform circular motion experiment, if the time for (20) oscillation is 9.28 s, then the angular frequency (ω) in (rad/s) is:

- A) 0.116 B) 15.6 C) 14.7 D) 13.53

5) If the periodic time for pendulum (1) is (τ_m) = 0.08s then (t_{15}) for The pendulum (2):

- A) 15s B) 0.12 s C) 18.s D) 0.18 s



6) In the simple pendulum experiment, if the length of the pendulum string increased four times, then the value of (τ) is:

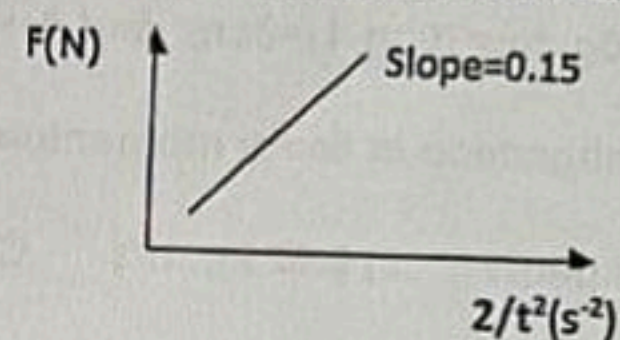
- A) Doubled B) tripled C) halved D) not affected

7) In a gravitational potential energy experiment, that consist of two masses ($M = 1\text{kg}$ on the surface of the air track and the hanging mass $m = 0.9\text{ kg}$) slide from rest, if the final kinetic energy of the system (7.23 J). What is the initial height of the mass (m)?

- A) 71cm B) 80 cm C) 13 cm D) 45 cm

8) In Newton's second law experiment, if ($M_{\text{total}}=200\text{g}$), the distance Which (M_w) moved in all trials, is:

- A) 70 cm B) 80 cm C) 75 cm D) 50 cm



9) In a gravitational potential energy experiment: when you fix (m) and change the initial height (h_i), the relation between (h_i) and the time (t) is:

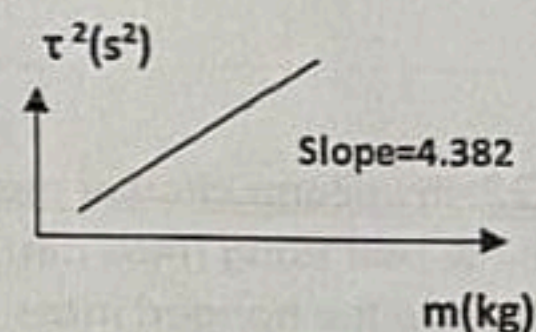
- A) Inversely proportional B) Directly proportional C) no relation

10) In Newton's second law experiment, when the force is constant the relation between the hanged mass (m_h) and the traveled distance (S) by (M_w) is:

- A) Inversely proportional B) Directly proportional C) no relation

11) In Hooks law experiment, the spring constant (K) is:

- A) 6 N/m B) 8 N/m C) 9 N/m D) 7 N/m



12) In uniform circular motion experiment, a light string can support a stationary hanging load of mass 12 kg for keeping an object of mass $m = 3.00\text{ kg}$ attached to the same string to rotate horizontally on uniform circular motion in a circle of radius $r = 0.800\text{ m}$, what is the speeds of the object?

- A) 5.6m/s B) 64 m/s C) 16 m/s D) 8 m/s

13) An electric balance, when turned on, gives 0.00 grams. The uncertainty (error) in mass by this electric balance

- A) 0.01 g B) 0.001g C) 0.05g D) 0.005g

14) In the physics laboratory, a student had measured the width of wood sample, and he repeated the measurement five times, if the value of $[\sum_1^5 (\bar{x} - x)^2 = 0.0351]$ then the standard deviation of the mean of his measurements is:

- A) 0.0524 B) 0.117 C) 0.234 D) 0.094

15) ($\text{Kg.m}^2/\text{s}^2$) is unit of:

- A) force B) momentum C) energy D) spring constant

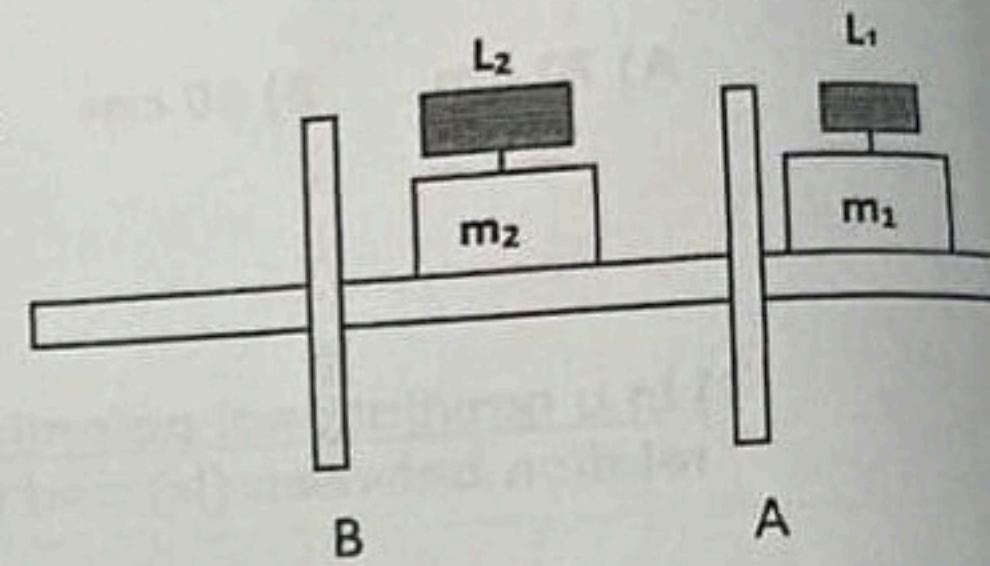
16) Using significant figures rules, find ($4. \times 6.45$):

- A) 26. B) 25.8 C) 26.0 D) 3×10^1

* In collisions experiment, according to the figure shown, answer (Q17), (Q18), (Q19):
For perfectly inelastic collision, (Where A and B are photogates) m_1 collides m_2 which is in static state.
If $m_1=200\text{g}$, $m_2=100\text{g}$, $L_1=3\text{cm}$, and $A_1=0.04\text{s}$.

17) The magnitude of linear momentum for m_1 and m_2 after collision (P_f) is:

- A) 0.15 kg.m/s B) 0.02 kg.m/s C) 0.03 kg.m/s D) 0.04 kg.m/s



18) In the previous collision, we can say:

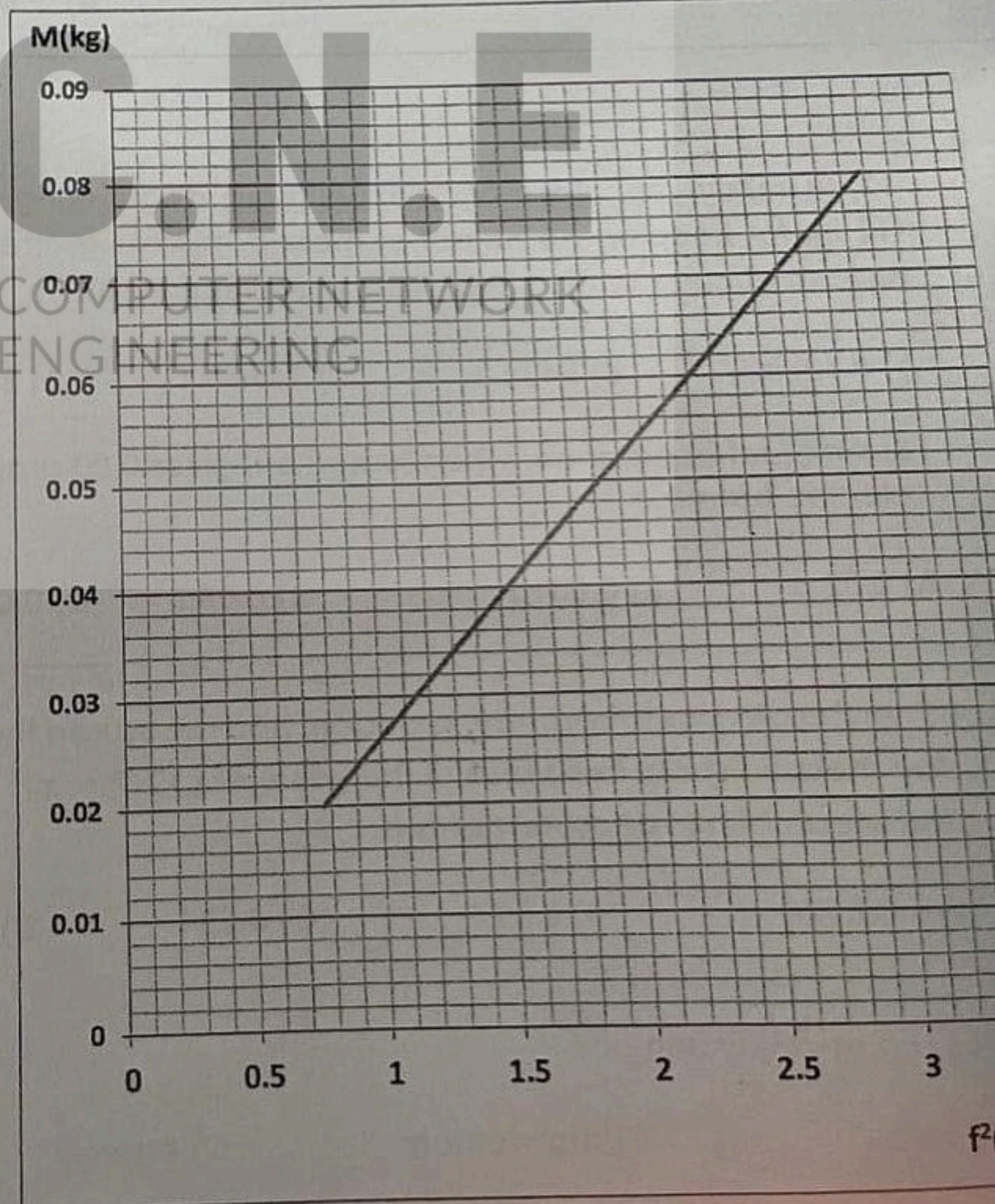
- A) $V_{1f} = V_{2f}$ B) $K_i \neq K_f$ C) $P_i = P_f$ D) all of them

19) If ($B_2=0.06\text{ s}$), ($B_1=0.15\text{ s}$). The length of the flag (L_2), which is positioned on (m_2) is:

- A) 6 cm B) 7.5 cm C) 4 cm D) 5 cm

Q2: In uniform circular motion experiment: in the part fixing ($r=50\text{ cm}$), ($m=15\text{g}$), and change the hanging mass (M), we got the graph shown. Answer the following blanks, and then move your answers to the table: (12 marks)

- 1) The physical meaning of slope
(Slope= -----).
- 2) The magnitude of the slope
(Slope= -----).
- 3) The magnitude of acceleration of gravity
($g=$ -----).
- 4) At ($T=0.60858\text{s}$), the magnitude of, the tension in rope ($T=$ -----).



*** Good luck ***